

to reach a well-defined goal. In other words, the process that we have inferred from Chapman's theory is an algorithm.

## ***Solving Problems with Algorithms***

The word *algorithm* came from the name of the great al-Khwarizmi, mentioned earlier. It's not an easy word to define, not least because its accepted definition has changed over time. Stated simply, an algorithm is just a set of instructions that produce a well-defined outcome. This is a broad definition; as we saw in the Introduction, tax forms and recipes for parfaits could rightly be considered algorithms.

Chapman's ball-catching process, or Chapman's algorithm as we may want to call it, is arguably even more algorithm-like than a recipe for a parfait, because it contains a looping structure in which small steps are taken repeatedly until a definite condition is reached. This is a common algorithmic structure you'll see throughout this book.

Chapman proposed an algorithmic solution to the outfielder problem because a solve-for- $x$  solution was not plausible (outfielders often don't know the relevant equations). In general, algorithms are most useful when the solve-for- $x$  strategy fails. Sometimes we don't know the right equations to use, but more often there is no equation that could fully describe a situation, the equation is impossible to solve, or we face time or space constraints. Algorithms exist at the edge of what is possible, and every time an algorithm is created or improved, we push the frontier of efficiency and knowledge out a little further.

Today, there is a common perception that algorithms are difficult, esoteric, mysterious, and strictly mathematical and that they require years of study to understand. The way our education system is structured today, we begin teaching children the solve-for- $x$  strategy as early as possible, and we explicitly teach algorithms only at the college or graduate school levels, if at all. For many students, it takes years to master the solve-for- $x$  strategy, and it always feels unnatural to them. People who have had this experience may assume that algorithms will feel just as unnatural, and will also be more difficult to understand because they are more "advanced."

However, the lesson I take from Chapman's algorithm is that we have gotten it all exactly backward. During recess, students learn and perfect